

On the Stability of Anisotropic Grain Boundaries

Caius Zielinski, Kenji Ueckermann, Lucia Oyelaran, Beatriz Castellanos, Eero Bhattacharya

Meridian Institute of Technology

Section: Materials

Published: 2025-06-28

DOI: 10.5281/mjas.14.2.06

Keywords: surrogate modelling; ablation study; noise robustness

Abstract

Work in materials research has long relied on assumptions we test directly here. The reduced model retains most of the explanatory power at a fraction of the computational cost. Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting.

1. Introduction

The reduced model retains most of the explanatory power at a fraction of the computational cost. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The measurements reported here were collected over twelve independent trials under controlled conditions. Sensitivity to the smoothing window proved modest across the range we examined. Calibration drift over the trial period remained below the threshold set out in the protocol. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

Sensitivity to the smoothing window proved modest across the range we examined. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Neither of the alternative formulations improved on the simple additive model in our setting.

2. Related Work

Replicate agreement was high, suggesting the protocol is robust to small handling differences. We release the processing scripts and intermediate artefacts to support independent verification. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Longer observation windows narrow the confidence interval but do not shift the central estimate. The reduced model retains most of the explanatory power at a fraction of the computational cost. Calibration drift over the trial period remained below the threshold set out in the protocol. Sensitivity to the smoothing window proved modest across the range we examined. The measurements reported here were collected over twelve independent trials under controlled conditions.

Replicate agreement was high, suggesting the protocol is robust to small handling differences. We release the processing scripts and intermediate artefacts to support independent verification. Residual error grows

approximately linearly with the number of coupled parameters. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

Replicate agreement was high, suggesting the protocol is robust to small handling differences. Taken together, these findings argue for treating the two regimes separately in future analyses. The reduced model retains most of the explanatory power at a fraction of the computational cost. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Sensitivity to the smoothing window proved modest across the range we examined. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The measurements reported here were collected over twelve independent trials under controlled conditions. Longer observation windows narrow the confidence interval but do not shift the central estimate.

The reduced model retains most of the explanatory power at a fraction of the computational cost. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We release the processing scripts and intermediate artefacts to support independent verification. The measurements reported here were collected over twelve independent trials under controlled conditions. Longer observation windows narrow the confidence interval but do not shift the central estimate. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

Calibration drift over the trial period remained below the threshold set out in the protocol. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Replicate agreement was high, suggesting the protocol is robust to small handling differences. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Taken together, these findings argue for treating the two regimes separately in future analyses. The reduced model retains most of the explanatory power at a fraction of the computational cost. Sensitivity to the smoothing window proved modest across the range we examined. Longer observation windows narrow the confidence interval but do not shift the central estimate.

3. Materials and Methods

We release the processing scripts and intermediate artefacts to support independent verification. The reduced model retains most of the explanatory power at a fraction of the computational cost. Neither of the alternative formulations improved on the simple additive model in our setting. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Sensitivity to the smoothing window proved modest across the range we examined. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The measurements reported here were collected over twelve independent trials under controlled conditions. Residual error grows approximately linearly with the number of coupled parameters.

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Sensitivity to the smoothing window proved modest across the range we examined. Longer observation windows narrow the confidence interval but do not shift the central estimate. Taken together, these findings argue for treating the two regimes separately in future analyses. The reduced model retains most of the explanatory power at a fraction of the computational cost. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The reduced model retains most of the explanatory power at a fraction of the computational cost. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The measurements reported here were collected over twelve independent trials under controlled conditions.

Residual error grows approximately linearly with the number of coupled parameters. The measurements reported here were collected over twelve independent trials under controlled conditions. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Longer observation windows narrow the confidence interval but do not shift the central estimate. Taken together, these findings argue for treating the two regimes separately in future analyses. Calibration drift over the trial period remained below the threshold set out in the protocol.

4. Experimental Setup

Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Calibration drift over the trial period remained below the threshold set out in the protocol. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Sensitivity to the smoothing window proved modest across the range we examined. The measurements reported here were collected over twelve independent trials under controlled conditions. The reduced model retains most of the explanatory power at a fraction of the computational cost. Neither of the alternative formulations improved on the simple additive model in our setting. We release the processing scripts and intermediate artefacts to support independent verification.

A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. The reduced model retains most of the explanatory power at a fraction of the computational cost. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Sensitivity to the smoothing window proved modest across the range we examined. Neither of the alternative formulations improved on the simple additive model in our setting. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Longer observation windows narrow the confidence interval but do not shift the central estimate.

The reduced model retains most of the explanatory power at a fraction of the computational cost. Longer observation windows narrow the confidence interval but do not shift the central estimate. Taken together, these findings argue for treating the two regimes separately in future analyses. Calibration drift over the trial period remained below the threshold set out in the protocol. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

Sensitivity to the smoothing window proved modest across the range we examined. The reduced model retains most of the explanatory power at a fraction of the computational cost. We release the processing scripts and intermediate artefacts to support independent verification. Residual error grows approximately linearly with the number of coupled parameters. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Longer observation windows narrow the confidence interval but do not shift the central estimate. Calibration drift over the trial period remained below the threshold set out in the protocol.

A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The reduced model retains most of the explanatory power at a fraction of the computational cost. Taken together, these findings argue for treating the two regimes separately in future analyses.